

**FATIMA JINNAH WOMEN UNIVERSITY**



**ARTIFICIAL INTELLIGENCE**

**PROJECT REPORT**

**TOPIC: TRAFFIC SIGNAL SYSTEM**

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# **Traffic Signal Simulation Using Artificial Intelligence:**

This Python program simulates the behavior of a traffic light system that uses artificial intelligence to make decisions based on real-time traffic conditions. Below is a complete breakdown of the code and its output, including its functioning and key features.

## **1. Purpose**

The goal of the program is to simulate a traffic control system with AI-based decision-making. The system determines whether the traffic signal for a specific direction should be **GREEN** or **RED** based on the number of cars waiting at each direction. This aims to improve traffic flow and reduce waiting times for vehicles.

## **2. Code Summary**

The program consists of two main classes:

1. **TrafficSignal:** Represents an individual traffic light for a specific direction.
2. **TrafficSystem:** Manages multiple traffic signals and coordinates their operation over several cycles.

### **Class: TrafficSignal**

- **Attributes:**
  - direction: Direction of the signal (e.g., "North-South" or "East-West").
  - state: Current state of the signal (RED or GREEN).
  - car\_queue: Number of cars currently waiting at the signal.
- **Methods:**
  - simulate\_traffic(): Randomly generates the number of cars waiting at the signal.
  - decide\_signal\_state(): Determines whether the signal should be GREEN or RED based on the number of cars waiting. If car\_queue > 5, the signal is set to GREEN.
  - display\_signal\_state(): Prints the current state of the signal and the number of cars waiting.

### **Class: TrafficSystem**

- **Attributes:**
  - Two instances of TrafficSignal for "North-South" and "East-West" directions.
- **Methods:**

- `simulate_traffic_system(cycles)`: Simulates the operation of the traffic light system over a specified number of cycles. For each cycle:
  - Traffic conditions are simulated for both signals.
  - The state of each signal is decided based on traffic conditions.
  - The current state and car queue for each signal are displayed.

### 3. Working Mechanism

#### 1. Simulation:

- The system simulates the arrival of cars for both traffic signals using random numbers (`car_queue`).
- Based on the number of cars, the signal's state is updated:
  - If cars > 5 → GREEN
  - Otherwise → RED

#### 2. Cycle Management:

- Each cycle represents a unit of time during which the system evaluates the current state of traffic, updates signals, and prints the result.

#### 3. AI-based Decision:

- AI techniques are used in the form of condition-based logic for decision-making. The system evaluates real-time data (cars waiting) to decide the signal states dynamically.

### 4. Output Analysis

The sample output provided is for 5 cycles. Let us analyze it:

#### Cycle 1:

- **North-South Signal:** RED with 5 cars waiting.
- **East-West Signal:** GREEN with 6 cars waiting.
- Decision: East-West has more cars, so it's set to GREEN while North-South is RED.

#### Cycle 2:

- **North-South Signal:** RED with 5 cars waiting.
- **East-West Signal:** RED with 1 car waiting.
- Decision: Both directions have low traffic, so both are set to RED.

### Cycle 3:

- **North-South Signal:** RED with 4 cars waiting.
- **East-West Signal:** RED with 3 cars waiting.
- **Decision:** Both directions have insufficient traffic to warrant a GREEN light.

### Cycle 4:

- **North-South Signal:** GREEN with 7 cars waiting.
- **East-West Signal:** GREEN with 8 cars waiting.
- **Decision:** High traffic in both directions, so both signals are GREEN to clear traffic.

### Cycle 5:

- **North-South Signal:** RED with 5 cars waiting.
- **East-West Signal:** RED with 3 cars waiting.
- **Decision:** Low traffic in both directions, so both are RED.

## 5. Artificial Intelligence in the Code

Although this code does not use advanced AI techniques like machine learning, it demonstrates foundational AI concepts:

- **Real-time Decision-Making:** The program evaluates traffic conditions dynamically to make decisions about the signal state.
- **Rules-Based System:** The decision-making is based on predefined rules (e.g., cars > 5 → GREEN).
- **Automation:** The system operates autonomously, requiring no manual intervention.

## 6. Recommendations for Improvement

To make the system more sophisticated and realistic:

### 1. Incorporate Fuzzy Logic:

- Instead of hard thresholds like "cars > 5," use fuzzy logic to handle uncertainty and make more nuanced decisions.
- Example: If cars = 4.5 (low to medium), the signal could be partially GREEN for a shorter duration.

### 2. Add Dynamic Timing:

- Adjust the duration of GREEN lights based on traffic density in each direction.

- Heavier traffic → Longer GREEN time.

### 3. Introduce Machine Learning:

- Train the system using historical traffic data to predict optimal signal timings.
- Use reinforcement learning (e.g., Q-Learning) for better decision-making over time.

### 4. Handle Simultaneous Signals:

- Implement rules to prevent simultaneous GREEN signals for intersecting directions to avoid accidents.

### 5. Visual Representation:

- Use visualization tools (e.g., matplotlib) to display the state of signals graphically, improving user understanding.

## 7. Applications

This simulation represents a simplified model of real-world traffic control systems. Such systems have numerous applications:

- **Smart Cities:** Adaptive traffic signals that reduce congestion and improve traffic flow.
- **Autonomous Vehicles:** AI-driven signals that interact with autonomous cars for efficient navigation.
- **Environmental Impact:** Reduced idle time at signals decreases fuel consumption and emissions.

## 8. Conclusion

The program successfully simulates a basic traffic light system using AI principles. While functional, its logic can be expanded to incorporate more advanced techniques like fuzzy logic or machine learning for greater efficiency and realism. This serves as an excellent foundation for developing intelligent traffic control systems for modern urban environments.